3.1.7 Oxidation, reduction and redox equations: Sheet 1

Oxidation states · 17 marks · about 25 minutes

Key information

- Oxidation is loss of electrons (oxidation state increases); reduction is gain of electrons (oxidation state decreases).
- Oxidising agent: electron acceptor (it is reduced). Reducing agent: electron donor (it is oxidised).
- Elements = 0. In compounds: F = −1; O = −2 (except peroxides −1, F_2O +2); H = +1 (except metal hydrides −1); Group 1 = +1; Group 2 = +2.
- Oxidation states in a compound add up to 0; in an ion they add up to the charge.

Section A: Self-quiz

01 Which compound contains chlorine in an oxidation state of +1?

[1 mark]

A Cl_2O
B $KClO_3$
C ClF_3
D CCl_4

02 Which compound contains a chlorine atom with an oxidation state of +4?

[1 mark]

A $KClO_4$
B CCl_4
C ClO_2
D ClO_2F

03 Which of these oxidation states is correct?

[1 mark]

A Chlorine in Cl_2 is −1
B Chromium in $K_2Cr_2O_7$ is +7
C Fluorine in F_2O is −1
D Hydrogen in NaH is +1

04 Determine the oxidation state of bromine and of fluorine in BrF_3

[1 mark]

05 Deduce the oxidation state of chlorine in $LiClO_4$

[1 mark]

Section B: Exam questions

06 Give the oxidation state of nitrogen in NO_2^- and in NO

[2 marks]

07 Deduce the oxidation state of nitrogen in NO_3^- and in NO

[2 marks]

08 Chlorine reacts with hot aqueous sodium hydroxide as shown in the equation.

$3\ Cl_2 + 6\ NaOH \rightarrow NaClO_3 + 5\ NaCl + 3\ H_2O$

08.1 Give the oxidation state of chlorine in $NaClO_3$ and in NaCl

[1 mark]

08.2 State, in terms of redox, what happens to chlorine in this reaction.

[1 mark]

09 Chlorine reacts with water: $Cl_2 + H_2O \rightleftharpoons HClO + HCl$

Deduce the oxidation state of chlorine in HClO and the oxidation state of chlorine in HCl

[1 mark]

10 Deduce the oxidation state of chromium in the $Cr_2O_7^{2-}$ ion.

[1 mark]

11 The reaction of calcium with water is a redox reaction. Explain, in terms of oxidation states, why this reaction involves both oxidation and reduction.

[2 marks]

12 Give an equation to show how magnesium is used as the reducing agent in the extraction of titanium from titanium(IV) chloride.

Explain, in terms of oxidation states, why magnesium is the reducing agent.

[2 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	**A**	Source: 7404/1 Nov 2020 Q21	1
02	**C**	Source: 7404/1 Nov 2021 Q16	1
03	**C**	Source: 7405/3 Jun 2022 Q10	1
04	Br (+)3, F −1	Source: 7405/1 Jun 2025 Q05.2 Both needed for the mark	1
05	(+)7	Source: 7405/1 Nov 2021 Q09.4 Accept VII	1
06	NO_2^-: +3 NO: +2	Source: 7404/1 Jun 2017 Q08.1 Allow III / 3 / 3+ and II / 2 / 2+	2
07	NO_3^-: (+)5 NO: (+)2	Source: 7405/1 Jun 2017 Q10.1 Allow Roman numerals	2
08.1	(+)5 and −1	Source: 7405/1 Jun 2019 Q05.3–05.4	1
08.2	Is oxidised and reduced	Allow undergoes disproportionation Allow gains and loses electrons	1
09	HClO = +1, HCl = −1	Source: 7404/1 Jun 2018 Q07.1 Both needed	1
10	+6 / VI / six / 6+	Source: 7404/1 Jun 2019 Q03.4	1
11	M1 Oxidation state of Ca increases (from 0 to +2), so Ca is oxidised M2 Oxidation state of H decreases, so H is reduced	Source: 7404/1 Jun 2024 Q02.3 If no marks awarded, then correct oxidation states of Ca and H before and after the reaction scores 1 mark	2
12	M1 $2Mg + TiCl_4 \rightarrow 2MgCl_2 + Ti$ M2 Mg changes oxidation state from 0 to +2 so electrons are lost / Ti changes oxidation state from +4 to 0, so gains electrons	Source: 7404/1 Nov 2020 Q05.2 Allow multiples Allow oxidation state of Mg increases so it is a reducing agent	2